XIAMEN UNIVERSITY MALAYSIA

ASSIGNMENT 2



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Try to solve the questions **Completely with all steps**! (Effective for grading)

The answer can be in **ONE PDF** format file only (individual activity)

Cheating in any condition will **consider zero Mark** in any exam and assignment!

only **1 Attempt** is allowed.

The deadline is the **23rd of December 2022 at 11:00 AM**

Late submission will impose **a penalty (communicate with me if delay happened)**!

Writing from solution manuals, web blogs, shared blogs, or any type of getting an answer will cause you zero marks!

Cheating and writing from **Internet resources** will consider as cheating, please avoid any kind of cheating! (**Do not share answers** at any cost!)

1. A parallel-plate, air-filled capacitor is being charged as in Figure A2.1. The circular plates have radius 4.00 cm, and at a particular instant the conduction current in the wires is 0.520 A.
(a) What is the displacement current density j_D in the air space between the plates?
(b) What is the rate at which the electric field between the plates is changing?
(c) What is the induced magnetic field between the plates at a distance of 2.00 cm from the axis?
(d) At 1.00 cm from the axis?

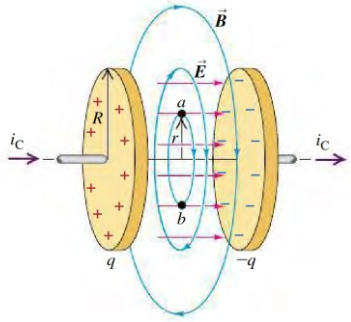


Figure A2.1

2. Three polarizing filters are stacked, with the polarizing axis of the second and third filters at 23.0° and 62.0°, respectively, to that of the first. If unpolarized light is incident on the stack, the light has intensity 55.0 W/cm^2 after it passes through the stack. If the incident intensity is kept constant but the second polarizer is removed, what is the intensity of the light after it has passed through the stack?

3. For the system of capacitors shown in Fig. A2.2, a potential difference of 45 V is maintained across ab.

(a) What is the equivalent capacitance of this system between a and b?

(b) How much charge is stored by this system?

(c) What is the potential difference across the total capacitors?

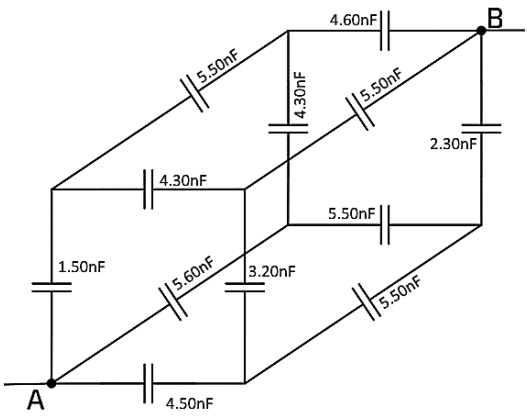


Figure A2.2

4. An aluminium wire with a diameter of 0.100 mm has a uniform electric field of 0.200 V/m imposed along its entire length. The temperature of the wire is 50.0°C. Assume one free electron per atom (T=20°C).

(a) Use the information in Table A2.1 to determine the resistivity of aluminium at this temperature.

(b) What is the current density in the wire?

(c) What is the total current in the wire?

(d) What is the drift speed of the conduction electrons?

(e) What potential difference must exist between the ends of a 2.00-m length of the wire to produce the stated electric field?

Table A2.1. Resistivities and Temperature Coefficients of Resistivity for Various Materials

Material	Resistivity[a] ($\Omega \cdot m$)	Temperature Coefficient[b] α $[(°C)^{-1}]$
Silver	1.59×10^{-8}	3.8×10^{-3}
Copper	1.7×10^{-8}	3.9×10^{-3}
Gold	2.44×10^{-8}	3.4×10^{-3}
Aluminum	2.82×10^{-8}	3.9×10^{-3}
Tungsten	5.6×10^{-8}	4.5×10^{-3}
Iron	10×10^{-8}	5.0×10^{-3}
Platinum	11×10^{-8}	3.92×10^{-3}
Lead	22×10^{-8}	3.9×10^{-3}
Nichrome[c]	1.00×10^{-6}	0.4×10^{-3}
Carbon	3.5×10^{-5}	-0.5×10^{-3}
Germanium	0.46	-48×10^{-3}
Silicon[d]	2.3×10^{3}	-75×10^{-3}
Glass	10^{10} to 10^{14}	
Hard rubber	$\sim 10^{13}$	
Sulfur	10^{15}	
Quartz (fused)	75×10^{16}	

5. The loop of wire shown in Figure. A2.3 forms a right triangle and carries a current I = 5.00 A in the direction shown. The loop is in a uniform magnetic field that has magnitude B = 3.00 T and the same direction as the current inside PQ of the loop.

(a) Find the force exerted by the magnetic field on each side of the triangle. If the force is not zero, specify its direction.

(b) What is the net force on the loop?

(c) The loop is pivoted about an axis that lies alongside PR. Use the forces calculated in part *(a)* to calculate the torque on each side of the loop.

(d) What is the magnitude of the net torque on the loop? Calculate the net torque from the torques calculated in part (c) Do these two results agree?

(e) Is the net torque directed to rotate point Q into the plane of the figure or out of the plane of the figure?

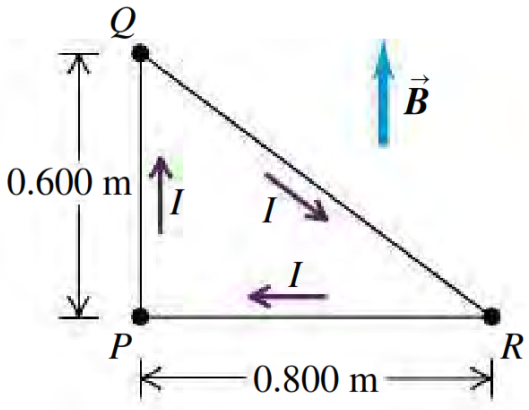


Figure A2.3

6. Rail guns have been suggested for launching projectiles into space without chemical rockets. A tabletop model rail gun (Figure A2.4) consists of two long, parallel, horizontal rails, l= 3.50 cm apart, bridged by a bar of mass m= 3.00 g that is free to slide without friction. The rails and bar have low electric resistance, and the current is limited to a constant I = 24.0 A by a power supply that is far to the left of the figure, so it has no magnetic effect on the bar. Figure A2.4 shows the bar at rest at the midpoint of the rails at the moment the current is established. We wish to find the speed with which the bar leaves the rails after being released from the midpoint of the rails.

(a) Find the magnitude of the magnetic field at a distance of 1.75 cm from a single long wire carrying a current of 2.40 A.

(b) For purposes of evaluating the magnetic field, model the rails as infinitely long. Using the result of part (a), find the magnitude and direction of the magnetic field at the midpoint of the bar.

(c) Argue that this value of the field will be the same at all positions of the bar to the right of the midpoint of the rails. At other points along the bar, the field is in the same direction as at the midpoint, but is larger in magnitude. Assume the average effective magnetic field along the bar is five times larger than the field at the midpoint. With this assumption, find

(d) the magnitude and

(e) the direction of the force on the bar.

(f) Is the bar properly modelled as a particle under constant acceleration?

(g) Find the velocity of the bar after it has travelled a distance d= 130 cm to the end of the rails.

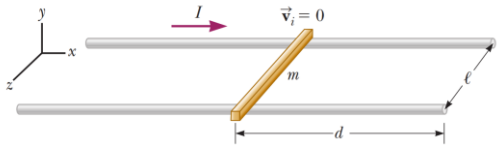


Figure A2.4

7. A 5.50-kg black cat and her four black kittens, each with mass 0.800 kg, sleep snuggled together on a mat on a cool night, with their bodies forming a hemisphere. Assume the hemisphere has a surface temperature of 31.0°C, an emissivity of 0.970, and a uniform density of 990 kg/m^3. Find

(a) the radius of the hemisphere,

(b) the area of its curved surface,

(c) the radiated power emitted by the cats at their curved surface, and

(d) the intensity of radiation at this surface. You may think of the emitted electromagnetic wave as having a single predominant frequency. Find

(e) the amplitude of the electric field in the electromagnetic wave just outside the surface of the cozy pile and

(f) the amplitude of the magnetic field.

(g) The next night, the kittens all sleep alone, curling up into separate hemispheres like their mother. Find the total radiated power of the family. (For simplicity, ignore the cats' absorption of radiation from the environment.)

8. In a darkened room, a burning candle is placed 1.50 m from a white wall. A lens is placed between the candle and the wall at a location that causes a larger, inverted image to form on the wall. When the lens is in this position, the object distance is p_1. When the lens is moved 90.0 cm toward the wall, another image of the candle is formed on the wall. From this information, we wish to find p_1 and the focal length of the lens.

(a) From the lens equation for the first position of the lens, write an equation relating the focal length f of the lens to the object distance p_1, with no other variables in the equation.

(b) From the lens equation for the second position of the lens, write another equation relating the focal length f of the lens to the object distance p_1.

(c) Solve the equations in parts (a) and (b) simultaneously to find p_1.

(d) Use the value in part (c) to find the focal length *f* of the lens.

9. Dinesh study about the mirrors. He wanted to locate one mirror in the room like Figure A2.5 with an overhead view of a room of square floor area and side *L*. At the center of the room is a mirror set in a vertical plane and rotating on a vertical shaft at angular speed ω about an axis coming out of the page. A bright red laser beam enters from the center point on one wall of the room and strikes the mirror. As the mirror rotates, the reflected laser beam creates a red spot sweeping across the walls of the room.

(a) When the spot of light on the wall is at distance *x* from point *O*, what is its speed?

(b) What value of *x* corresponds to the minimum value for the speed?

(c) What is the minimum value for the speed?

(d) What is the maximum speed of the spot on the wall?

(e) In what time interval does the spot change from its minimum to its maximum speed?

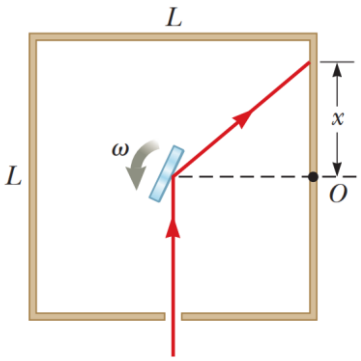


Figure A2.5

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